

Diego Cardenas-Wallenfelt

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EDUCATION

California Polytechnic University,
Junior, B.S. Mechanical Engineering

Pomona, CA

GPA: 3.50

Expected Graduation Year: 2028

ENGINEERING WORK EXPERIENCE

Vulcan Wireless — Mechanical Engineering Intern

Carlsbad, CA | May 2026 – August 2026

- Designed mechanical components, assemblies, and test fixtures in SolidWorks, producing detailed engineering drawings for fabrication and supporting hardware through design, prototyping, testing, and iteration.
- Designed, fabricated, and validated a PPA-CF vibration test fixture, achieving a 39% mass reduction vs. CNC aluminum; developed and executed 3 g, 10–3000 Hz sine-sweep and 10 GRMS random-vibration tests, analyzing transmissibility and modal response to validate the printed fixture for acceptance testing.
- Developed thermal models from drawings, BOMs, component specifications, and material properties; conducted SolidWorks thermal analysis and thermal-vacuum testing to evaluate thermal-interface solutions and correlate experimental results with predictions.
- Fabricated RF antenna-chamber fixtures, selecting ESD-safe and magnetically compatible materials to satisfy test requirements while minimizing interference with sensitive RF measurements.
- Conducted NOVICE Total Ionizing Dose (TID) analysis of PCB components and aluminum chassis shielding to support radiation-hardening decisions.

ENGINEERING PROJECTS

- Formula SAE – Cal Poly Pomona Clutch & Shifter Captain | 2024 – Present:**
 - Led end-to-end design and analysis of the clutch and shifter subsystems using SolidWorks and FEA to validate structural performance, packaging, and manufacturability.
 - Developed an electronic clutch system with programmable actuation profiles, enabling adjustable bite-point control and improved launch consistency for competition events.
 - Engineered a pneumatic shifting system, performing cylinder sizing, valve selection, pressure calculations, and response time to achieve fast and repeatable gear changes.
- Liquid Rocket Club (Turbine Design) 2024:** Designed and simulated a liquid rocket nozzle and radial-inflow turbine using NASA CEA and SolidWorks CFD; analyzed flow properties, pressure ratios, efficiency, and NsDs parameters to optimize propulsion performance.
- Arduino/PCB Device:** Designed and built a custom electromechanical device integrating a KiCad PCB, C++ firmware, user I/O, and CAD-designed FDM enclosure.

RELEVANT COURSEWORK

Thermodynamics • Fluid Mechanics • Dynamics • Materials • Engineering Graphics/CAD • Computer Aided Computations (MATLAB) • Electromagnetism & Circuit Analysis • Stress Analysis

TECHNICAL SKILLS

- Technical Skills:**
 - CAD & Analysis:** SolidWorks (CAD, Drawings, FEA, Modal/Vibration, Thermal, CFD), Onshape
 - Testing & Prototyping:** Vibration, Shock, Thermal-Vacuum, FDM Additive Manufacturing, Test Fixture Design
 - Electronics & Programming:** MATLAB, C++, Arduino, KiCad, PCB Design
 - Other:** NOVICE Radiation Analysis

INTERESTS

- Interests:** Running (state-level 4×800 finalist), Music, Making/Prototyping.